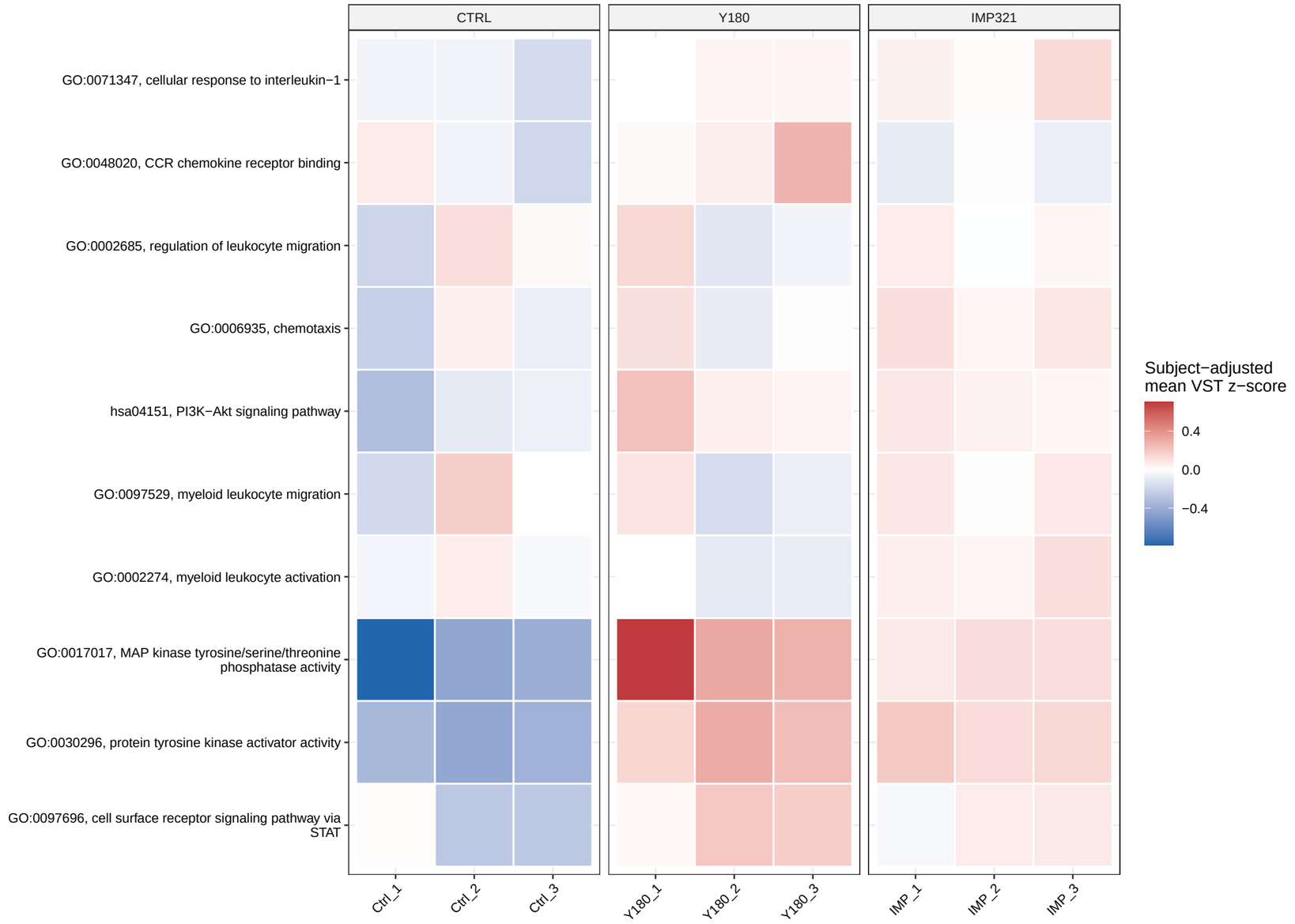
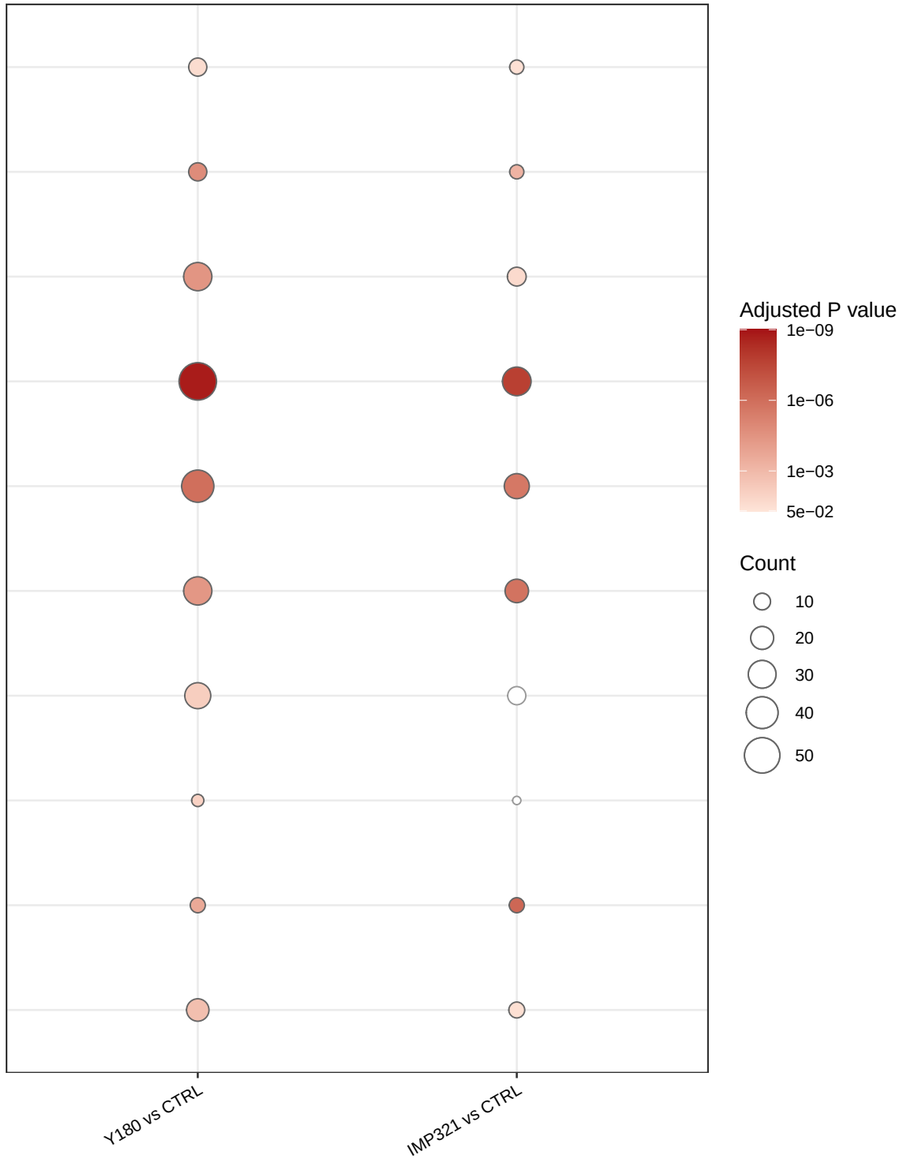
A | Full−pathway expression in all paired samples



All expressed annotated members; subject effects removed with ~subject + group, retaining group effects.
Each column is one paired biological sample; expression remains descriptive and does not imply activation.

B | Treatment−versus−control UP ORA



Count = pathway−overlapping UP DEGs. Filled = ORA FDR < 0.05.
BH correction covers all eligible GO/KEGG sets and both contrasts.